

Moosa Abbasi

813-729-2005 | moosaabbasi@usf.edu | linkedin.com/in/moosaabbasi | github.com/moosaabbasi | moosaabbasi.com

Education

University of South Florida, Honors College

Bachelor of Science in Computer Science

GPA: 3.86/4.00 | Green and Gold Presidential Scholar, Dean's List

Tampa, FL

Aug. 2023 – May. 2027

Relevant Coursework: Data Structures, Analysis of Algorithms, Computer Architecture, Software Engineering, Computer Logic & Design, Automata Theory, Introduction to Artificial Intelligence, Discrete Structures, Honors Capstone

Technical Skills

Languages & Databases: Python, C, C++, JavaScript, SQL, PHP, DynamoDB, MySQL

Frameworks & Libraries: Streamlit, React.js, Flask, OpenCV, Pandas, NumPy, Plotly, scikit-learn, XGBoost, Prophet

Cloud & DevOps: AWS (Lambda, API Gateway, S3, IAM), Google Cloud (Cloud Run, Secret Manager, GCS), Docker, Git, GitHub Actions

Security & Testing: Burp Suite, OWASP ZAP, OWASP WSTG v4.2, Playwright, HTTP Traffic Analysis, XSS/SQLi, Prompt Injection

Certifications: Machine Learning Specialization (Supervised ML, Advanced Learning Algorithms, Unsupervised Learning): Andrew Ng, DeepLearning.AI & Stanford University, Coursera · 2026

Experience

Undergraduate Research Assistant: AI Course Companion (AICC)

May 2026 – Present

University of South Florida | Dr. Oguzhan Topsakal

- Built an **automated security testing suite** using **Python + Playwright** to assess AICC across **OWASP WSTG v4.2** categories and **LLM prompt injection testing**, integrating a **Gemini-based evaluator** to reduce false positives against the LLM backend.
- Conducted **black-box and gray-box security testing** of the **OAuth flow and backend APIs**, tracing authentication and data-serving logic across the codebase and compiling an **OWASP-mapped, severity-rated security report** for the supervising faculty.
- Containerized the security-testing infrastructure with **Docker** and deployed it to **Google Cloud Run** using **Secret Manager, Cloud Storage, and IAM**, creating a **cloud-hosted platform** for ongoing AICC security evaluation.

Student Software Developer: AI Course Companion (AICC)

Aug 2026 – Present

University of South Florida

- Remediated **security vulnerabilities** identified through the AICC security assessment, shipping production fixes across **session management, data exposure, security headers, CORS controls, and Google Cloud Storage access**.
- Developed and maintained **PHP 8.1 backend** and **JavaScript frontend** components across the AICC codebase, implementing **API endpoints, application logic, LLM integrations**, and data-serving workflows for the Canvas-embedded platform.
- Built **admin dashboard features** for **usage and student-question analytics**, developing supporting **backend APIs and frontend modules** to surface application insights and recurring student queries, while resolving **production issues**.

Project Management Intern

May 2024 – Aug 2024

Huawei Technologies

- Resolved visibility gaps across **5+ concurrent DMD projects** by building structured **tracking dashboards** that standardized reporting cycles and cross-team workflows, improving on-time delivery by **20%**.
- Eliminated siloed communication by introducing a **data-driven reporting** system and standardized **project documentation** workflow, reducing delays by **25%** and boosting team efficiency by **15%**.

Projects

Smart Job Tracker

May 2025 – Present

AWS Lambda, DynamoDB, API Gateway, Cognito, SES, Chrome Extension (MV3), Streamlit, Python

Tampa, FL

- Eliminated manual logging via a **Chrome Extension (MV3)** with a **3-layer detection system** using **MAIN world XHR monkey-patching** and **MutationObserver**, plus a **chrome.alarms** queue guaranteeing **zero application loss** on service worker suspension.
- Engineered a **5-Lambda serverless backend** with **DynamoDB user-scoped isolation**, **API Gateway REST API** protected by a **Cognito JWT** authorizer, **preemptive token refresh** with race-condition locking, and automatic **401 retry** across all 3 clients.
- Deployed a live **Streamlit** dashboard with **EventBridge + SES** daily email reminders for application tracking; reduced total operating cost to **\$0/month** by architecting entirely within the **AWS Free Tier**.

USF Emissions Analysis Dashboard

Jan 2026 – Present

Python, Streamlit, Plotly, Pandas, scikit-learn, XGBoost, Prophet, GHG Protocol

Tampa, FL

- Quantified **~83.8M kg CO₂** across **5 categories** (Scope 1 + Scope 3 Cat 4/6/7/9) by building a full-stack **GHG emissions dashboard** in **Python/Streamlit**, applying **GHG Protocol** and **EPA SmartWay** emission factors to **~50,000 synthetic records**.
- Built **3 ML models**: **IsolationForest** anomaly detection flagging **500 shipments** with **6.3M kg CO₂** savings potential, **Prophet** 24-month emissions forecasting, and **XGBoost** regression at **R²=0.9965** with invoice amount as the top emission predictor.
- Deployed on **Streamlit Cloud** via **GitHub** with automated redeployment; presented findings to an airport industry client in collaboration with **Patel's College of Global Sustainability** for Scope 1 and Scope 3 strategy recommendations.